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Seizure Medications

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Continuing Education Activity

The mainstay treatment strategy for seizures is medication management. However, much like the prescription of any other pharmaceutical agent, a clinician must balance efficacy with adverse events, and provide consideration for cost, drug interactions, patient preference, and availability. This activity outlines the indications, mechanisms of action, methods of administration, important adverse effects, contraindications, and monitoring, of various seizure medications, so providers can direct patient therapy in treating indicated disorders as part of the interprofessional team with seizure medications, with a basis on the current knowledge for optimal utilization.

Objectives:

- Identify the various classes of anti-seizure medications and their mechanisms of action.
- Review the indications for therapy initiation by seizure drug subclass.
- Describe the contraindications and adverse effects of the various agents in the seizure medication classification.
- Explain the importance of improving care coordination among the interprofessional team to enhance the delivery of care for patients who can benefit from therapy with seizure medications.

Access free multiple choice questions on this topic.

Indications

The mainstay treatment strategy for seizures is medication management. However, much like the prescription of any other pharmaceutical agent, a clinician must balance efficacy with adverse events, while considering cost, drug interactions, patient preference, and availability. This article is intended to provide a general overview of seizure medications and the current knowledge base for optimal utilization.[1][2][3]

The American Academy of Neurology (AAN) and the American Epilepsy Society offered recent guidelines on the initiation of seizure medications after a first unprovoked seizure in an adult. The focus is predominantly on an individualized approach, with patient autonomy at the forefront. Before making any decisions, factors that may increase risk should be identified and include abnormal brain imaging, abnormal electroencephalogram (EEG), and the presence of nocturnal seizures. Patients should understand that the risk of seizure recurrence is greatest within the first 2 years (21% to 45%) and that this risk may be mitigated with the initiation of medications. However, adverse events of anti-seizure medications, though most frequently mild and reversible, should be reviewed and discussed in detail.

Of note, certain seizure medications have been used off-label for a variety of other indications, including but not limited to:

- Anxiolytics: Pregabalin, clonazepam, clobazam
- Migraine relief: Zonisamide, valproic acid, topiramate
- Mood stabilizer: Valproic Acid, lamotrigine, carbamazepine
- Neuropathic pain relief: Pregabalin, gabapentin, carbamazepine
- Weight loss: Zonisamide, topiramate
- Antiparkinsonian agent: Zonisamide

Mechanism of Action

Anti-epileptic drugs (AEDs) are numerous. There are a variety of mechanisms of action, and some AEDs possess multiple mechanisms of action. Although precise mechanisms of some drugs remain elusive, anti-seizure medications tend to be grouped by their principal mode of action. A brief review of some of the major drugs on the market is provided below.

Some AEDs act on the sodium channels by either blocking their repetitive activation (phenytoin, carbamazepine) or by enhancing their slow inactivation (lacosamide). Others work on calcium channels by blocking either T-type calcium channels (ethosuximide, valproic acid) or the N- and L-type calcium channels (zonisamide). Lamotrigine works by blocking sodium channels, blocking N- and L-type calcium channel, and modulating H-current. Topiramate works by blocking sodium channels, alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA) receptors, and by inhibiting carbonic anhydrase. Other mechanisms through which AEDs act are by enhancing gamma-aminobutyric acid (GABA)-A receptors (phenobarbital, benzodiazepines), blocking N-methyl-D-aspartic acid (NMDA) receptors (felbamate), and opening neuronal potassium channels (ezogabine).[4] More detailed descriptions of some of these drugs are provided in the following paragraphs.

Carbamazepine (CBZ) is used for focal and generalized seizures. It is also used for patients with bipolar disorder and trigeminal neuralgia. By inhibiting voltage-gated sodium channels, CBZ reduces neuronal firing. It is metabolized by the cytochrome P450 system and is a strong inducer of these hepatic enzymes. Common side effects include gastrointestinal upset, hyponatremia, rash, itch, drowsiness, dizziness, blurred vision, and headache. Rare but dangerous side effects include Stevens-Johnson syndrome, toxic epidermal necrolysis, leukopenia, and aplastic anemia (pancytopenia).

Oxcarbazepine is structurally similar to CBZ and thus wields a similar mechanism of action. It is also comparable to CBZ in terms of efficacy for focal and secondarily generalized tonic-clonic seizures. Patients may experience dizziness, headache, ataxia, nausea, rash, double vision, and/or hyponatremia. The latter effect is thought to be secondary to the syndrome of inappropriate secretion of antidiuretic hormore (SIADH) and may occur in 25-75% of patients.

Phenytoin is one of the oldest anti-seizure medications and is still widely used for focal and generalized seizures. It is also administered for status epilepticus. In addition, practitioners may invoke phenytoin as a second-line agent for patients with mixed seizure types (e.g., tonic-clonic and myoclonic). As mentioned, phenytoin blocks voltage-gated sodium channels, but other possible mechanisms revolve around decreased synaptic transmission, smaller changes in ionic gradients involving the sodium-potassium ATPase pump, and inhibition of calcium-calmodulin phosphorylation. Of note, phenytoin is a broad spectrum inducer of the CYP system and, therefore, may reduce the effectiveness of many forms of hormonal oral contraception (OCP) medications. Major adverse effects include gingival hyperplasia, body hair increase, folic acid depletion, rash, and worsening bone density. Patients should also be counseled regarding the possible development of confusion, ataxia, double vision, and neuropathy with long-term usage.

Lacosamide stabilizes hyperexcitable membranes and inhibits repetitive neural firing via the slow inactivation of voltage-gated sodium channels. Another possible mechanism may involve binding to collapsin response mediator protein 2 (CRMP2), which has been implicated in epileptogenesis. Lacosamide is used as monotherapy or adjunct therapy for focal-onset seizures in patients aged 4 years and up. It tends to be tolerated decently well, with dizziness, ataxia, and nausea being the most typically encountered adverse effects. Importantly, dose-dependent PR interval prolongation on electrocardiogram (EKG) is a well known effect, and so EKG monitoring is recommended alongside initiation of therapy.

Phenobarbital is another very old drug. It is used for generalized and focal seizures, but its sedating properties make it difficult to use in the modern clinical realm. Regarding mechanism, phenobarbital is a barbituate and binds to the GABA-A receptor, prolonging its open state to allow for more chloride influx and consequent cellular hyperpolarization. It is a potent inducer of the CYP system and thus can decrease serum levels of other drugs metablized by this pathway.

Vigabatrin is an inhibitor of an enzyme that metabolizes GABA (GABA transaminase), thereby elevating GABA concentrations within the central nervous system. Vigabatrin is used adjunctively for refractory focal seizures, but it can also be used as monotherapy in this setting. It is also useful in the pediatric world for infantile spasms, particularly in children with tuberous sclerosis complex. Patients should be precautioned regarding possible vision loss. The drug can also lead to MRI abnormalities (but without clear neurologic deficits), fatigue, dizziness, and headache.

Topiramate has several modes of action, including blockage of voltage-gated sodium channels, GABA transmission enhancement, and NMDA receptor antagonization. It also blocks carbonic anhydrase to a modest extent. Topiramate is used as monotherapy in both pediatric and adult medicine for focal-onset and primary generalized tonic-clonic seizures (ages 10 and up). The drug can also be used as adjunctive therapy for focal seizures or in patients with Lennox-Gastaut syndrome (ages 2 and up). Patients may commonly experience weight loss, impaired cognition, and difficulties with expressive language. Some patients may experience fatigue, depression, headache, and/or paresthesias. Metabolic acidosis with tachypnea and/or calcium phosphate nephrolithiasis (kidney stones) may also occur.

Valproate (valproic acid, VPA) is another broad spectrum medication used to treat both focal and generalized seizures. Similar to topiramate, it possesses multiple mechanisms of action, including voltage-gated sodium channel inhibition, augmentation of GABA concentrations, and mild inhibition of T-type calcium currents (but not to the extent of ethosuximide). Valproate is metabolized by the liver and is an inhibitor of CYP enzymes. It should be used with caution in patients with hepatic insufficiency and should be avoided altogether in patients with urea cycle disorders due to high risk of hyperammonemia. Parenthetically, combining topiramate and valproate can lead to increased levels of ammonia. This combination should be closely monitored if encountered in clinical practice. With that said, topiramate is not alone in this regard; other carbonic anhydrase inhibitors, enzyme-inducing anti-seizure drugs, and antipsychotic medications can increase the risk of hyperammonemic encephalopathy in the setting of VPA use. Other general side effects of VPA include nausea/vomiting, tremor, easy bruising, weight gain/insulin resistance, metabolic syndrome, and subclinical hypothyroidism. Rarely, VPA can cause acute hepatocellular injury with jaundice or acute pancreatitis. VPA should be avoided during pregnancy, as teratogenic neurotoxicity has been observed.

Levetiracetam's precise mechanism is not clear, but studies have linked its effectiveness to the binding of synaptic vesicle protein 2A (SV2A). It encompasses a very broad spectrum in terms of its efficacy and is used as adjunctive therapy for focal-onset seizures in children and adults, myoclonic seizures in patients with juvenile myoclonic epilepsy (ages 12 and up), and primary generalized tonic-clonic seizures (ages 6 and up) in patients with idiopathic generalized epilepsy. The drug is uniquely metabolized independent of the hepatic CYP system, and it does not induce CYP enzymes like phenytoin, phenobarbital, and carbamazepine. Importantly, levetiracetam does not necessitate a prolonged titration period like other drugs (e.g., lamotrigine with its risk of Stevens-Johnson syndrome).

Levetiracetam is extremely well-tolerated, with common adverse effects revolving around behavioral disturbances, drowsiness, dizziness, and upper respiratory infections.

In summary, it is now abundantly clear that anti-seizure medications wield disparate mechanistic profiles, but they all effectively suppress epileptic seizures in one way or another. Accordingly, grouping the drugs together by mechanism is a very helpful organizing principle. From this viewpoint, it may become easier to appreciate that some drugs have different efficacy profiles for different seizures types and epilepsy syndromes. Ethosuximide is an exception with its specific limited use with respect to absence seizures, but almost all of the other available medications listed above have potential use against generalized and focal seizures. [5]

Administration

Seizure Types

After the diagnosis of epilepsy, choosing of an AED is largely dependent on the classification of the seizure type. In the most general sense, seizures can be classified as either partial or generalized. Partial seizures can be further broken down into those that do not affect awareness (simple partial) or those that do (complex partial).

Simple partial seizures affect focal areas of the brain that, in turn, cause focal neurological findings like loss of motor function in one limb, sensory dysfunction in a specific body region, or changes to vision or speech, without affecting awareness of the event. Complex partial seizures are similar but do include brief moments of confusion that quickly resolve once the seizure has ended.

Generalized seizures affect the entire body and can exist as tonic-clonic seizures (where muscles stiffen initially, which is followed by jerking and spasming of the body), absence seizures (where individuals may stare off into space, and there is a brief lapse in awareness, often confused with daydreaming), and atonic seizures (where there is a sudden loss of muscle tone, frequently resulting in falls).

The International League Against Epilepsy (ILAE) recommends that epilepsy be diagnosed when any of the following exist: a history of two unprovoked seizures at least 24 hours apart, or an unprovoked seizure when the risk for a subsequent seizure is greater than 60% after two unproved seizures over the subsequent 10 years, or seizures that are part of an epilepsy syndrome.

Initiation of Antiepileptic Drugs

AEDs can be broken down into two categories: broad-spectrum and narrow-spectrum. Broad-spectrum AEDs treat a wide variety of seizure types, as the name suggests, and are a good initial choice, especially when the classification of seizure type is uncertain. These AEDs include but are not limited to levetiracetam, lamotrigine, zonisamide, topiramate, valproic acid, clonazepam, perampanel, clobazam, and rufinamide.

Narrow spectrum AEDs primarily are for the treatment of focal or partial seizures. These include but are not limited to lacosamide, pregabalin, gabapentin, carbamazepine, oxcarbazepine, ezogabine, phenytoin, and vigabatrin.

Monotherapy is the ideal pathway for the treatment of seizures, but newer AEDs have had difficulty obtaining FDA approval as a monotherapy agent due to the difficulty of achieving approval requirements. However, anecdotally, and by examining the current evidence base, second-generation AEDs appear to be an appropriate choice, as they have demonstrated similar efficacy when compared to older AEDs and may be better tolerated.

One large randomized trial, the Standard and New Antiepileptic Drugs (SANAD) trial, demonstrated some comparative advantages of certain AEDs when treating focal or generalized epilepsy. In the end, when comparing valproate, lamotrigine, or topiramate for generalized seizures, they recommended valproic acid as their first-line choice. Additionally, when comparing carbamazepine, gabapentin, lamotrigine, oxcarbazepine, and topiramate for focal seizures, lamotrigine was cited as the first-line choice.

Of note, regarding partial onset seizures specifically, perampanel, lacosamide, brivaracetam, and eslicarbazepine acetate are a few of the recently FDA-approved seizure medications for monotherapy.

Discontinuation

When to discontinue AEDs is less clear. In children, consensus recommendations appear to be after two years of remaining seizure free. However, in adults, some research has estimated the risk of seizure recurrence immediately after cessation of AEDs at the 2-year mark at 30%. The risk appears to diminish with time if an individual continues without further seizures. Taking this into account, it is apparent why guidelines in adults are more conservative and recommend a seizure-free period of 2 to 5 years while advising against driving for three months after AED cessation.

Adverse Effects

Patients should be taught that adverse events with anti-seizure medications, according to the AAN, may range from 7% to 31%, but are mostly mild and reversible. In general, it is important to review the specific medication insert for a complete list of adverse events.[6][7][8]

Of the more mild and common side effects, patients should be advised to monitor for headaches, fatigue, dizziness, blurry vision, nausea, weight gain or loss, mood disorders, and neurocognitive problems. The potential for allergic reactions is present amongst all medications and should be monitored upon initiation.

With chronic use, many AEDs carry a side effect of osteoporosis, and general recommendations are to supplement diets with calcium and vitamin D, while encouraging routine exercise habits.

Of the more serious side effects, Stevens-Johnson syndrome, agranulocytosis, aplastic anemia, hepatic failure, pancytopenia, multiorgan hypersensitivity, psychosis, and lupus syndrome have all been reported. Although the risk for suicidality is low, the FDA has required all AEDs to carry a suicide warning.

Also of concern are drug-drug interactions, which occur most commonly with the older generation of AEDs, as they can affect hepatic enzymes (e.g., the cytochrome P450 system). If these hepatic enzymes are induced, it can lead to the rapid metabolism of other medications, potentially leading to their subtherapeutic levels in the body. If, on the other hand, hepatic enzymes are inhibited, toxic levels of medications can be reached due to impaired breakdown through the liver. For example, carbamazepine, phenytoin, and phenobarbital are liver enzyme inducers and can increase the metabolism of concomitantly prescribed drugs like warfarin. In contrast, valproate is a liver enzyme inhibitor.

Both inducing and inhibiting medications can raise significant concern when managing certain comorbid conditions where therapeutic doses of medications are of the utmost importance, including HIV, cancer, endocrine disorders, and cardiovascular disease.

Contraindications

A contraindication to most medications, in general, is a prior history of hypersensitivity or allergic reaction to that medication. Other contraindications exist but are more drug-specific, including hepatic failure, certain blood diseases, narrow-angle glaucoma, and familial short QT syndrome, to name a few.

Valproic acid and felbamate are associated with hepatotoxicity and, therefore, contraindicated in patients with hepatic failure. The liver metabolizes many others, and dosage adjustments need to be made before initiation.

Similarly, those with renal impairments need to adjust dosing when AEDs are primarily excreted via the kidneys. Adjustments can be made using the glomerular filtration rate. Hemodialysis also may require dosing adjustment to ensure the medication remains in the respective therapeutic range.

Likewise, because of variability in metabolism and excretion rates, caution must be exercised in children and the elderly.

Regarding pregnancy, guidelines proffered by the AAN in 2009 report that women should remain on their current AED, as switching medications can increase the risk of breakthrough seizure, and adding another AED increases the risk of congenital disabilities. Folic acid should be supplemented for any woman of childbearing age, especially those considering becoming pregnant. If possible, optimization of the medication regimen, by identifying the lowest effective dose with the least amount of medications, is ideal. Women who experience their first seizure during pregnancy should follow the same management strategy as if they were not pregnant.

Through the duration of the pregnancy, serum drug levels should regularly be monitored, and dosages titrated or adjusted to within therapeutic ranges.

Unfortunately, there is insufficient research into comparative efficacy and teratogenicity for recommendations of specific AEDs in pregnancy. However, valproic acid deserves special mention as it is contraindicated in pregnancy due to consistent evidence of teratogenicity, including dose-dependent effects on IQ, adaptive functioning, increased risk of ADHD, and increased risk of major congenital malformations.

As benefits are believed to outweigh risks, the consensus remains that AEDs can continue to be taken while breastfeeding. However, measurable levels can be found in breastmilk, and available research has not determined what risk this poses to newborns.[9][10][11]

Monitoring

Once administered, monitoring of serum medication levels is recommended to establish a therapeutic baseline and assess for toxicity. If the patient remains stable, these levels can be checked yearly, along with a complete blood count, a comprehensive metabolic panel, and liver function testing.

Toxicity

Toxic symptoms in AEDs are idiosyncratic and are believed to occur more frequently in the first-generation AEDs. One such example is carbamazepine, a commonly implicated first-generation AED in acute or chronic toxicity. Possible symptoms may include ataxia, dystonia, sinus tachycardia, hyperthermia, coma, arrhythmias, respiratory depression, and death. Another example is valproic acid, which can have toxic effects that include metabolic and hematological disruption, pancreatitis, central nervous system (CNS) depression, optic nerve atrophy, respiratory depression, cardiopulmonary arrest, brain edema, and coma.[12][13][14]

Treatment can range from supportive measures to high-flow hemodialysis, plasmapheresis, or charcoal hemoperfusion.

Patients should be educated before initiating any AED to alert treating physicians when adverse events are encountered. It is also important to note that therapeutic reference ranges are considered guidelines, and levels outside of the range do not necessarily indicate toxicity or a need to adjust dosing. However, if a given clinical picture correlates with elevated serum drug levels, then, indeed, treatment is warranted.

Enhancing Healthcare Team Outcomes

Once a patient has been started on anti-seizure medications, education of the patient is vital. In addition to the prescribing practitioner, both the patient's nurse and pharmacist should educate the patient on adherence on medication compliance. Further, injury is not uncommon during a seizure, and the family has to be educated about seizure precautions surrounding activities like driving and swimming. Patients with a diagnosis of seizures should not undertake unsupervised activities in the water or work at heights. Depending on the frequency of seizures, some patients may not be allowed to drive. Because most anti-seizure medications have side effects, the patient must be told

to follow up with the primary caregiver and be monitored closely. Finally, patients should be cautioned against changing the dose frequency or timing, without first speaking to the healthcare provider. [15][16][17](Level V)

Review Questions

- Access free multiple choice questions on this topic.
- Comment on this article.

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